



POWER CURVE

S144 3.0 / 3.15 MW

Variant :HTV light | 50 Hz
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List of changes*

Revision	Date	Change aspect
00	2022-06-17	New document
01	2023-02-03	Editorial changes
02	2023-03-28	Nameplate updated to S144-3.0 /3.15 MW Rated wind speed updated

* Changes to the design of the wind turbine generator (WTG) from the standard detailed in this specification have to be made as follows:

Buyer originated changes

This specification can be revised by Specification Change Notices (SCN) which describe the changes to be made to the specification.

Manufacturer originated changes

This specification can be revised by the seller without the buyer's consent to incorporate development changes if such changes do not adversely affect price, delivery, guaranteed performance of the WTG, or the interchangeability or replaceability requirements of this specification.

Development changes

Development changes are changes considered necessary to improve the WTG, prevent delays or to ensure compliance with the purchase agreement. The seller has to notify the buyer of all changes that are made as described in this section, by giving the buyer revised pages for this specification on a periodic basis

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1 **General information**

Item	Parameter
Turbine Model	S144 3.0/3.15 MW
Wind class	IEC S
Rotor diameter	144 m
Rotor blade	SB70
Rated power	3.0/3.15 MW
Cut in wind speed	3.0 m/s
Rated wind speed	9.2/9.35 m/s (without turbulence intensity according to IEC 61400)
Cut out wind speed	18 m/s
Grid Frequency	50Hz

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2 Power Curve

S144 power curve at standard Air Density (AD) of 1.225 kg/m³.

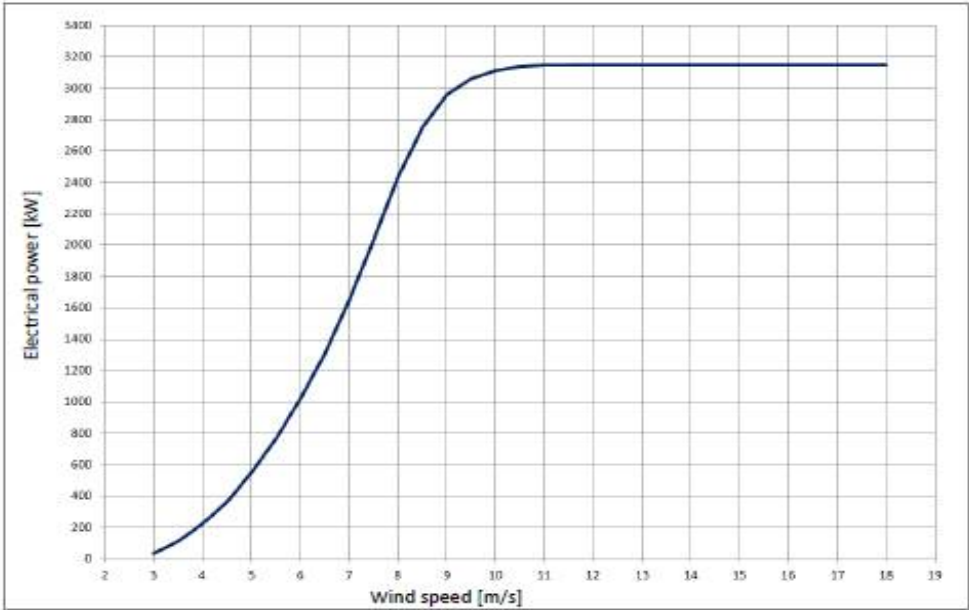


Figure 1: S144 Power curve [Air Density (AD) = 1.225 kg/ m³]

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Table 1: Power Curve for S144 [Air Density (AD) = 1.225 kg / m³]

Wind speed [m /s]	Electrical power [kW]
3.0	35
3.5	111
4.0	222
4.5	366
5.0	554
5.5	768
6.0	1019
6.5	1306
7.0	1642
7.5	2023
8.0	2429
8.5	2751
9.0	2958
9.5	3064
10.0	3113
10.5	3139
11.0	3150
11.5	3150
12.0	3150
12.5	3150
13.0	3150
13.5	3150
14.0	3150
14.5	3150
15.0	3150
15.5	3150
16.0	3150
16.5	3150
17.0	3150
17.5	3150
18.0	3150

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3 Disclaimer

The herewith specified standard power curve is valid under the following conditions:

- The power curve verification/test shall be in accordance to the SuzlonStandard power curve verification procedure with reference to IEC /3/
- The rotor blades have to be according to the nominal blade design conditions (for example no presence of ice, snow, no damage or dirt).
- The wind turbine is operating in its specified range.
- Special operating conditions for example low temperature variant (LTV) or noise reduced mode are not considered.
- The wind turbine is operating without being influenced by the park controller.
- The electrical power shall be measured between the Out-put terminal of the Turbine Power panel and the Low Voltage terminal of the Turbine Transformer.

4 **References**

- /1/ General -Specification - WSCC-2223-WTGS-14450_00
- /2/ IEC 61400-1 Wind turbines - Part 1: Design requirements
- /3/ IEC 61400-12-1:2005 Wind turbines - Power performance measurements of electricity producing wind turbines

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